

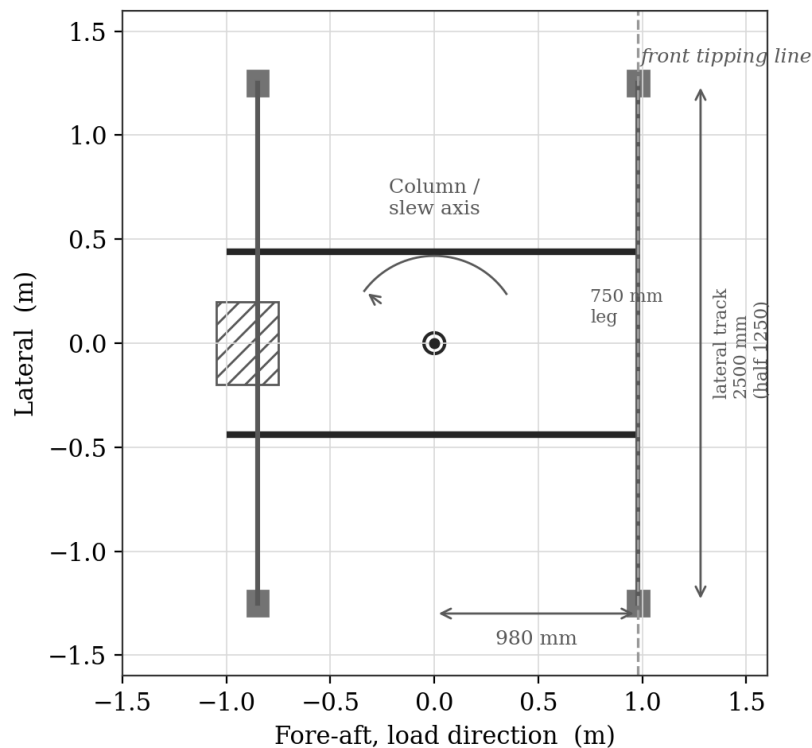
PROJECT 40467361

STRUCTURAL CALCULATION AND
LOAD VERIFICATION REPORT

1 Tonne Mobile Floor Jib Crane

Deployed-outrigger configuration: stability in the # arrangement and the leg-member check, on Q235 material

Deployed footprint, plan view: two rails plus four transverse legs (# arrangement)



Deployed footprint with the four legs in the # arrangement. The legs stand square to the rails, so they widen the lateral track to 2500 mm foot-to-foot while the front line stays 980 mm ahead of the column.

Field	Detail
Project reference	40467361
Configuration	Outriggers deployed (four transverse legs, # arrangement)
Material	Q235 carbon structural steel (fy 235 MPa)
Rated capacity (as marked)	1000 kg at 1400 mm reach
Document revision	Rev C, supersedes Rev B on the Q235 material and 2500 mm track
Date	12 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a, parametric script C2_v3

Document Control

This report assesses the 1 tonne mobile jib crane on the manufacturer drawing set with its four outrigger legs deployed. It is the companion to the wheels-only report, which covers the same crane standing on its wheels with the legs stowed. Both documents run off one parametric model and one input set. This revision, Rev C, replaces Rev B on two changes: the crane material is Q235, not S275, and the deployed lateral track is taken from the updated general-arrangement drawing at 2500 mm foot-to-foot. The material change lowers every stress-based factor of safety; the wider track lifts sideways stability slightly. Forward stability is set by the geometry and the masses, so it does not move with the material.

Revision history

Rev	Date	Description	Status
A	10 Jun 2026	First issue of the deployed-outrigger study, front line at the 1200 mm place-holder.	Superseded
B	11 Jun 2026	Front overhang corrected to 980 mm; forward-stability and leg-reaction figures revised.	Superseded
C	12 Jun 2026	Material set to Q235 (fy 235 MPa, was S275). Deployed track taken from the updated drawing at 2500 mm. Stress-based factors revised down; sideways stability revised up. Items above the slewing bearing recomputed on Q235.	Issued for review

Basis and limitations

- **Material.** The crane material is Q235 (Chinese carbon structural steel, GB/T 700), confirmed by the client. Nominal yield is 235 MPa; all sections are 6 mm wall, well under 16 mm, so the full 235 MPa applies with no thickness de-rating. Q235 is the equivalent of ASTM A36 and EN S235JR. The member allowable is $235 / 1.5 = 156.7$ MPa.
- **Drawings as issued.** Leg dimensions come from Base DETAIL-11, DETAIL B-B and DETAIL-12. The deployed track is read off the updated general-arrangement plan view at 2500 mm foot-to-foot; the inner frame width is dimensioned at 1100 mm. Section sizes and the front overhang come from the base and assembly sheets.
- **Front overhang.** The 980 mm front line follows from two readings that agree: the base top-view chain places the slew centre about 1076 mm behind the front edge, and the warning plate marks 420 mm at the front, so the front line sits at 1400 minus 420, or 980 mm. The new drawing keeps the 2360 mm wheelbase, so this is unchanged.
- **Deployed geometry.** The legs are taken square to the rails at 90 degrees. The deployed feet land at the 2500 mm track shown on the drawing; the as-built deployed width should be confirmed on the machine.
- **Scope of physics.** Static lifting only. Stability is a moment balance on static loads. Wind, side pull and snatch are outside this scope.
- **Not a design.** This verifies an existing unit against its rated chart in the deployed state. It does not replace load testing or a competent-person examination, which are carried out separately with the approval officer before the crane is put to work.

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1. Executive Summary

This report checks the 1 tonne mobile jib crane covered by the wheels-only report, but with the four outrigger legs deployed and the crane material taken as Q235. The legs form a # arrangement: the two fore-aft rails that carry the wheels run front to back, and a front and a rear transverse beam cross them, each reaching out to either side to a foot. Seen from above, the rails and the two leg beams read as a #.

Two facts decide the result. First, the legs sit at 90 degrees to the rails, so they reach sideways and not forward. Putting them down is what lets the crane slew, because the wide lateral track keeps it from going over when the boom swings off the fore-aft line. They do nothing for forward tipping, because a transverse leg adds no reach in the luffing plane. The front tipping line stays at the front frame line, 980 mm ahead of the column. Second, the material is Q235, with a nominal yield of 235 MPa, about 15 percent below the 275 MPa assumed in the earlier revisions. Every stress-based check therefore falls, and three members that were at or above target on the stronger steel now sit below it.

Forward stability is set by the masses and the geometry, not the steel, so it is unchanged. The mid-reach point, 700 kg at 2320 mm, governs, and its factor against tipping is 0.96. A factor below one means the lifted-load moment about the front line already exceeds the restoring moment, so on a static balance the crane is past its forward tipping point at that chart position.

Sideways stability improves slightly on the wider 2500 mm track. The crane is steady close in, at a factor of 5.83 at 1400 mm, but it still tips at medium and long reach, where the factors are 1.17 and 1.42. Reaching 1.50 at the worst point needs the track opened to about 2784 mm, a little beyond the 2500 mm shown, so the deployed width must be measured on the machine and, if it adjusts, set wider before slewing at mid or long reach.

On Q235 the front-support members split. The deployed leg member, taken as a 750 mm cantilever in RHS 60 x 100 x 6 carrying half the front reaction, reaches 144 MPa for a factor of 1.64, and still passes. The base outrigger arm, which passed at 1.53 on S275, now falls to 1.30 and fails, because the weaker steel drops the allowable below the 180 MPa it carries. The leg passes; the outrigger arm does not.

Nothing above the slewing bearing is helped by the legs, and the weaker steel makes it worse. The column stays the governing structural item, now at a yield factor of 0.85, its base weld falls to 0.73, and the boom outer section, marginal at 1.48 on S275, drops to 1.26 and fails on Q235. Deploying the legs rescues none of these.

The conclusion follows directly. The legs make slewing possible and ease the leg member, but forward tipping is unchanged and below one at mid-reach, sideways stability is short until the track is confirmed wider, the outrigger arm and the boom outer now fail on the weaker steel, and the column and weld limits are untouched. The crane cannot be signed off to the full 1 tonne chart on a working-stress basis. The route to the structural items, the re-rate or the column upgrade, is the one set out in the companion report and it does not depend on the legs.

Headline results, deployed configuration (# arrangement, legs at 90 degrees), Q235

Item	Result	FoS	Target	Verdict
Forward tipping, worst (700 kg @ 2320 mm)	9.20 kN·m	0.96	1.50	Fail (tips)

Item	Result	FoS	Target	Verdict
Sideways tipping, worst (700 kg @ 2320 mm)	7.35 kN·m	1.17	1.50	Fail
Deployed leg member (RHS 60x100x6)	144 MPa	1.64	1.50	Pass
Base outrigger arm (Q235)	180 MPa	1.30	1.50	Fail
Boom outer section (Q235)	186 MPa	1.26	1.50	Fail
Column, combined N+M (Q235)	277 MPa	0.85	1.50	Fail
Column-to-base weld (Q235)	235 MPa	0.73	1.50	Fail
Lateral track provided	2500 mm	—	2784 mm	Short

Section 9 places the deployed and wheels-only results side by side. Sections 5, 6 and 7 give the stability balances and the leg-member calculation in full.

2. Introduction and Scope

2.1 Background

The unit is a mobile floor jib crane rated to lift 1 tonne. It runs on four wheels, carries a counterweight at the rear, and luffs a three-stage telescopic boom on a single hydraulic cylinder. The upper assembly slews on a bolted bearing flange above the base frame. The warning plate fitted to the machine tells the operator to put the legs down before slewing, so the rated chart is written for the legs-down state.

The wheels-only report covers the crane standing on its wheels with the legs stowed. This report is the companion study for the deployed state. It looks at what the four legs change and at what they leave alone, and it carries the Q235 material through every member.

2.2 What this report covers

The deployed legs act on the part of the crane below the slewing bearing, which is stability and the members that carry the load down to the ground. This report covers:

- Forward tipping in the luffing plane, taken about the front foot line
- Sideways tipping with the boom slewed across the rails, taken about a side foot line
- The track the legs need for sideways stability to reach the target, against the 2500 mm they give
- Bending in the deployed leg member, and the base outrigger arm it shares load with, on Q235
- A statement of which wheels-only results carry over, recomputed on the weaker steel

Everything above the slewing bearing, the column, the boom, the pins and the slewing bolts, is fixed by the boom and the lifted load and does not change with the legs. Those checks come from the wheels-only study and are summarised in Section 8 on the Q235 material rather than worked again here.

2.3 Configuration assessed

All stability results in this report are for the crane standing on its four deployed legs in the # arrangement, with the legs square to the rails at 90 degrees. The wheels-only results appear only for comparison. If the legs in fact splay forward, forward stability will be better than reported here; Section 11 flags this.

3. The Deployed Configuration

The crane base is a pair of fore-aft rails with a wheel at each corner. Stowed, that narrow rail track is the whole footprint, and it is the reason the crane is unsafe to slew on its wheels. Deploying the legs adds two transverse beams, one on the front wheel line and one toward the rear, each reaching out to both sides. The four feet at their ends, with the two rails, make the # seen in plan.

The leg dimensions come from Base DETAIL-11. Each leg arm is 750 mm long, with a 230 mm drop to an 80 mm foot pad, an M16 jack screw for levelling, and a Phi32 hinge pin. DETAIL B-B records a longer 798 mm arm and DETAIL-12 a Phi20 eye foot. The 750 mm arm of DETAIL-11 is taken as the leg-member reach. Because the beams sit square to the rails, the feet widen the footprint sideways only. The front foot stays level with the front frame line, 980 mm ahead of the column, so it adds nothing to the forward overhang.

The deployed lateral track is taken from the updated general-arrangement plan view, which dimensions the foot-to-foot width at 2500 mm. The inner frame width is dimensioned at 1100 mm. This 2500 mm track, with a half-track of 1250 mm, governs sideways stability and is carried through Section 6. It supersedes the 2376 mm that earlier revisions derived from the bare leg-arm projection.

Quantity	Value	Source
Leg arm (lateral reach)	750 mm	DETAIL-11
Foot drop	230 mm	DETAIL-11
Foot pad	80 mm	DETAIL-11
Levelling jack screw	M16 (Phi16)	DETAIL-11
Leg hinge pin	Phi32	DETAIL-11
Alternative leg arm	798 mm	DETAIL B-B
Inner frame width	1100 mm	GA plan view
Deployed lateral track	2500 mm (foot-to-foot)	GA plan view
Front foot ahead of column	980 mm (front frame line)	Base drawing + warning plate
Leg member section	RHS 60 x 100 x 6	drawing
Material	Q235 (fy 235 MPa)	client-confirmed

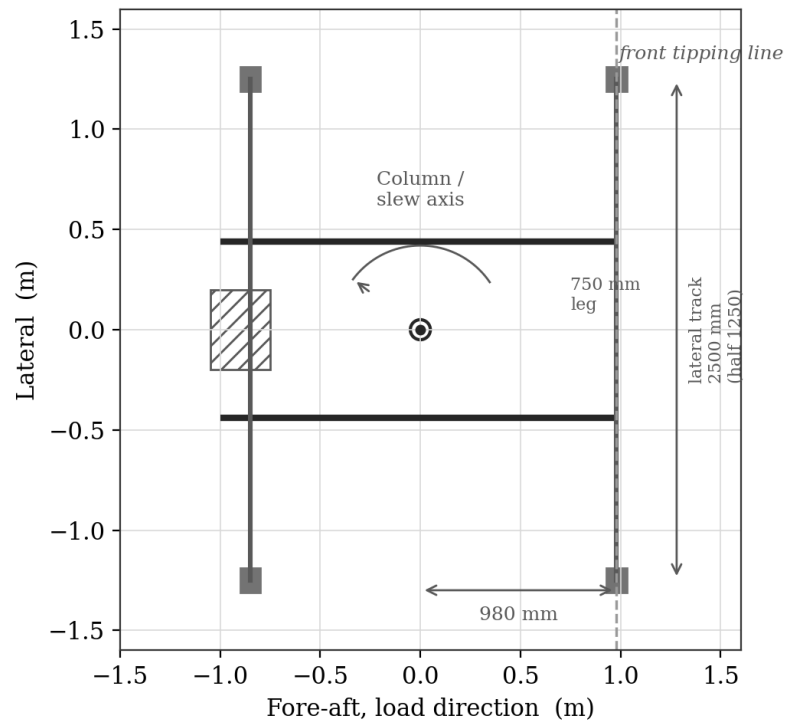
Deployed footprint, plan view: two rails plus four transverse legs (# arrangement)

Figure 3.1 Deployed footprint in plan. The legs give a 2500 mm foot-to-foot track, while the front foot stays on the 980 mm front line.

4. Materials and Load Model

All members are hollow steel sections in grade Q235, with a nominal yield of 235 MPa. Q235 is the Chinese carbon structural steel of GB/T 700, equivalent to ASTM A36 and EN S235JR, with an ultimate strength of 370 to 500 MPa; the minimum 370 MPa is used for weld checks. All sections are 6 mm wall, so the full 235 MPa yield applies with no thickness de-rating. The member allowable is the yield divided by the 1.5 target. Pins carry a 3.0 target, as is normal for lifting connections.

Allowable stresses (Q235, $f_y = 235$ MPa, $f_u = 370$ MPa)

Member bending / direct stress (target FoS 1.5):

$$\sigma_{\text{allow}} = f_y / 1.50 = 235 / 1.50 = 156.7 \text{ MPa}$$

Pin / bolt shear (target FoS 3.0):

$$\tau_{\text{allow}} = 0.6 \times f_y / 3.0 = 47.0 \text{ MPa}$$

Weld ($a = 6$ mm, target FoS 1.5):

$$f_w_{\text{allow}} = 0.7 \times f_u / 1.50 = 0.7 \times 370 / 1.50 = 172.7 \text{ MPa}$$

The rated chart gives three points, with capacity falling as reach grows. The factored load applies the 1.25 dynamic factor used for HC1 hoist duty.

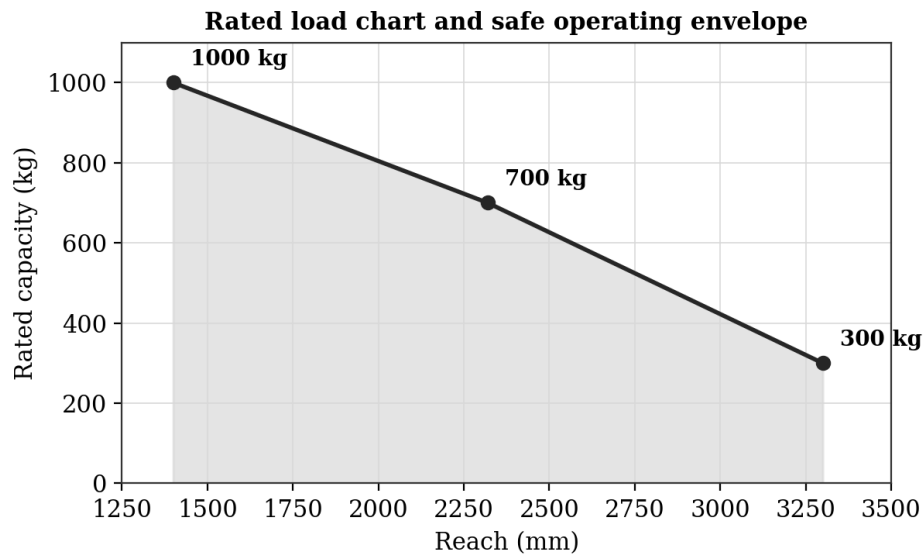


Figure 4.1 Rated load chart with the safe operating envelope shaded.

Chart point	Reach (mm)	Rated load (kg)	Static load (kN)	Factored load (kN)
Close	1400	1000	9.81	12.26
Mid	2320	700	6.87	8.58
Far	3300	300	2.94	3.68

The structural items carried over in Section 8 are worst at the mid-reach point, where the design moment at the column is 19.91 kN·m. Stability, treated next, is assessed on the static loads, because the 1.25 dynamic factor applies equally to the overturning and the restoring sides and cancels in the ratio.

5. Forward Stability

Forward tipping is a moment balance about the front foot line, 980 mm ahead of the column. With the legs at 90 degrees, that line sits in the same place as the front wheel, so the balance is identical to the wheels-only case. The restoring moment comes from the crane weight and counterweight behind the line, and the overturning moment from the lifted load ahead of it. The material does not enter a stability balance, so these numbers are the same on Q235 as on S275.

Overturning balance about the front foot line

$$\begin{aligned}\text{Restoring } M_R &= m_{\text{crane}} \times g \times (L_{\text{fwd}} + e_{\text{cg}}) \\ &= 700 \times 9.81 \times (0.980 + 0.300) = 8.79 \text{ kN.m} \\ \text{Overturning } M_{OT} &= m_{\text{load}} \times g \times (\text{reach} - L_{\text{fwd}}) \quad [\text{static load}]\end{aligned}$$

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	4.12	2.13	Pass
700 kg @ 2320 mm	9.20	0.96	Fail (tips)
300 kg @ 3300 mm	6.83	1.29	Fail

The mid-reach point governs at a factor of 0.96, below one and well below the 1.50 target. A factor under one means the overturning moment already beats the restoring moment, so on the static balance the crane is over its forward tipping point at that position. Transverse legs do not move the front line, so they cannot improve forward stability, and the weaker steel does not change it either.

6. Sideways Stability During Slew

Sideways tipping is the case the legs are there for. With the boom slewed across the rails, the crane can go over a side foot line set at the half-track, 1250 mm from the centreline on the 2500 mm deployed track. The restoring moment is the crane weight acting at that half-track. The overturning moment is the lifted load acting at its reach, less the half-track, since the foot line is offset toward the load by that amount. Like forward stability, this balance does not depend on the material.

Overturning balance about a side foot line

Restoring $M_R = m_{\text{crane}} \times g \times t_{\text{half}} = 700 \times 9.81 \times 1.250 = 8.58 \text{ kN.m}$

Overturning $M_{OT} = m_{\text{load}} \times g \times (\text{reach} - t_{\text{half}}) \quad [\text{static load}]$

Half-track from the drawing: $t_{\text{half}} = 2500 / 2 = 1250 \text{ mm}$ (track 2500 mm)

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	1.47	5.83	Pass
700 kg @ 2320 mm	7.35	1.17	Fail
300 kg @ 3300 mm	6.03	1.42	Fail

The close-in lift is steady, but the mid and far points both fall below target, at 1.17 and 1.42. The mid-reach point is again the worst, and it sets how wide the feet must stand. Solving the balance for a factor of 1.50 at that point gives a half-track of 1392 mm, a full track of 2784 mm. The 2500 mm drawn track is about 284 mm short of that, so the deployed width must be confirmed on the machine and, if the feet adjust, set to at least 2784 mm before the crane is slewed at mid or long reach. The wider 2500 mm track is an improvement on the 2376 mm of earlier revisions, but it does not by itself clear the target.

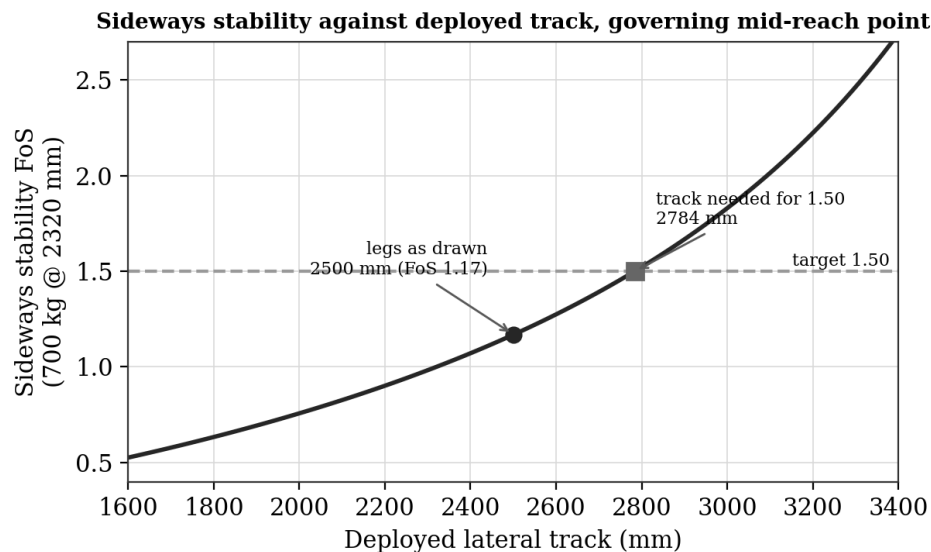


Figure 6.1 Sideways stability against the deployed track at the governing mid-reach point. The 2500 mm track from the drawing sits below the 1.50 target; about 2784 mm is needed.

7. The Deployed Leg Member

The deployed leg is checked as a cantilever picking up the vertical reaction at its foot. The largest foot reaction comes from the heaviest lift taken in close, not from the tipping case, because member force follows the vertical load while tipping follows the lever arm. The front pair carries 17.43 kN at 1000 kg and 1400 mm, so each leg arm carries about 8.72 kN at its 750 mm reach.

Deployed leg, RHS 60 x 100 x 6, cantilever (Q235)

front reaction (1000 kg @ 1400 mm) = 17.43 kN total -> 8.72 kN per leg

lever (leg arm) = 750 mm = 0.750 m

$M_{leg} = 8.72 \times 0.750 = 6.54 \text{ kN.m}$

$S_x = 45.5 \text{ cm}^3 = 45.5 \times 10^{-6} \text{ m}^3$

$\sigma = M_{leg} / S_x = 6.54 \times 10^3 / 45.5 \times 10^{-6} = 143.7 \text{ MPa}$

$FoS = f_y / \sigma = 235 / 143.7 = 1.64 \rightarrow \text{Pass}$

The leg passes at 1.64 on Q235, down from 1.91 on the stronger steel but still above target. The base outrigger arm, which carries the same front reaction over its fore-aft cantilever to the front wheel line, comes out at 180 MPa. On Q235 its factor is $235 / 180 = 1.30$, below the 1.50 target, so it now fails. On S275 the same 180 MPa gave 1.53 and passed. The leg member clears the weaker steel; the outrigger arm does not.

8. Items Carried Over from the Wheels-Only Check

The legs attach to the base frame, below the slewing bearing. The axial load and the bending moment that pass up through the bearing into the column, and on into the boom and its pins, are fixed by the boom geometry and the lifted load. None of them change when the base stands on legs instead of wheels. What does change from the earlier revisions is the material: every stress-based factor below is recomputed on Q235, and the pin factors fall with the lower steel shear allowable. The set is collected below.

Check	FoS	Target	Verdict
Boom outer (100x140x6)	1.26	1.50	Fail
Boom middle (80x120x6)	2.17	1.50	Pass
Boom inner (60x100x6)	2.64	1.50	Pass
Column, combined N+M	0.85	1.50	Fail
Column buckling	67.6	1.50	Pass
Base outrigger arm (Q235)	1.30	1.50	Fail
Pivot pin (Phi30)	2.08	3.00	Fail
Cylinder rod-end pin (Phi20 as modelled)	1.03	3.00	Fail
Hook pin (Phi20)	3.44	3.00	Pass
Slewing bolts (5 x M16, 8.8)	1.69	1.50	Pass
Weld, column base	0.73	1.50	Fail
Weld, boom bracket	2.92	1.50	Pass

The cylinder rod-end pin is listed at the conservative 20 mm diameter and fails on that basis. The only hinge pin shown near the cylinder mount is the Phi40 Base DETAIL-4 pin, which clears the check at 4.11 on

Q235; the as-built pin diameter should be confirmed on the machine. The boom outer and outrigger arm, which were marginal or passing on S275, now fail on the weaker steel. The slewing bolts and the hook pin are Grade 8.8 fasteners, so they do not move with the base-steel grade and stay as before. The column at 0.85 and its base weld at 0.73 remain the governing structural items in both configurations.

9. Stowed versus Deployed

The table below sets the two configurations side by side, both on Q235. The legs change the bottom of the load path and the stability picture. They leave the top of the structure exactly as it was; the weaker steel acts on both states alike.

Item	Wheels stowed	Legs deployed (# arrangement)
Front tipping line ahead of column	980 mm	980 mm (no forward reach)
Lateral foot track	876 mm	2500 mm
Forward tipping, worst point	FoS 0.96, Fail	FoS 0.96, Fail (unchanged)
Sideways tipping during slew	narrow track, tips	FoS 1.17, Fail (track short)
Front ground-support member	outrigger arm 180 MPa, FoS 1.30, Fail	leg arm 144 MPa, FoS 1.64, Pass
Column, combined N+M	277 MPa, FoS 0.85, Fail	unchanged
Column-to-base weld	235 MPa, FoS 0.73, Fail	unchanged
Boom outer section	186 MPa, FoS 1.26, Fail	unchanged
Pins and slewing bolts	Q235 base, Gr 8.8 bolts	unchanged

Read across, the pattern is consistent. The legs widen the track and ease the leg member, so sideways tipping becomes possible to manage and the leg passes. Forward tipping, the column, the weld, the boom and the outrigger arm are the same in both states, because the legs do not act on them and the material is common to both.

Stability with the legs deployed (# arrangement, 90 deg transverse)

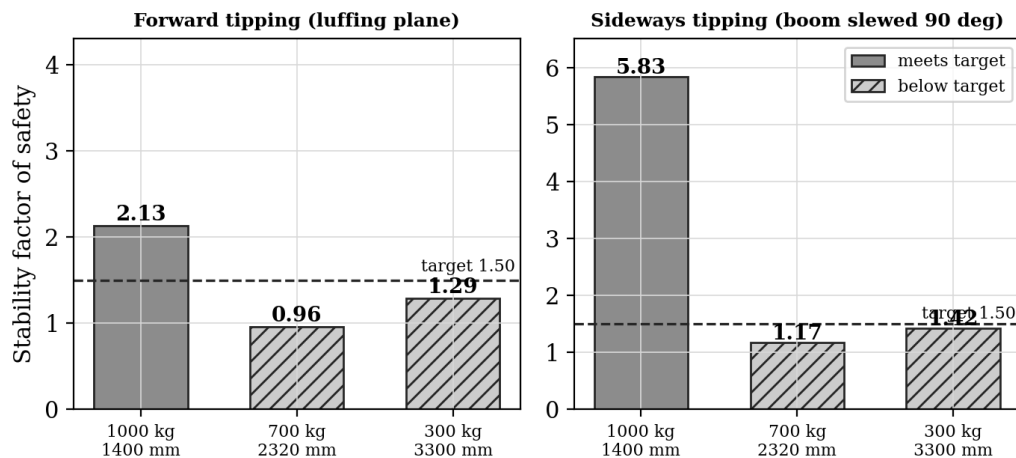


Figure 9.1 Stability in the deployed state on the 2500 mm track. Forward tipping is unchanged from the wheels-only case and sits below one at mid-reach; sideways tipping clears the target only close in.

The factor-of-safety summary below collects the member and stability checks on Q235. The hatched bars are below their required factor; the black tick on each bar marks the target.

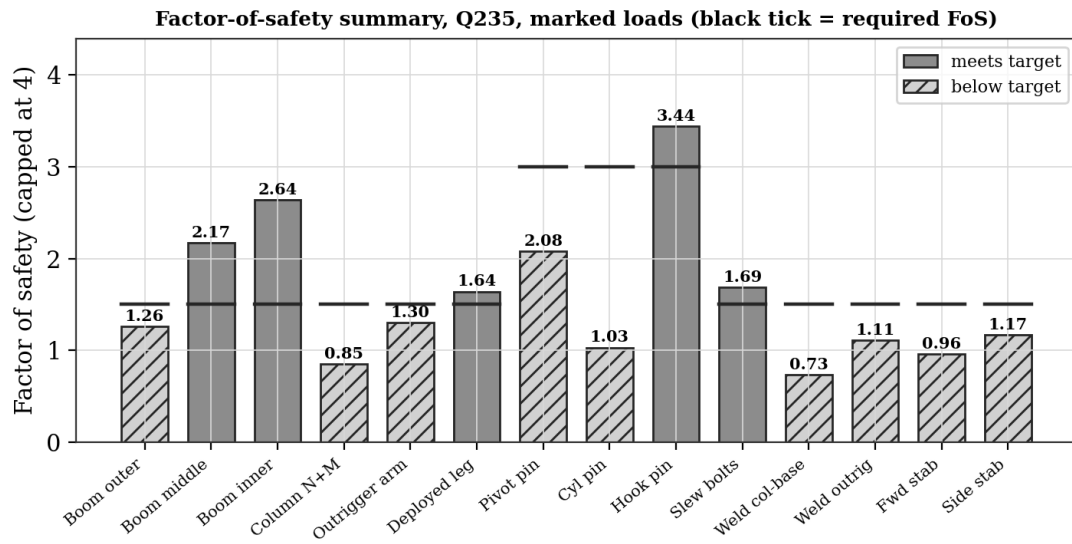


Figure 9.2 Factor-of-safety summary on Q235 at the marked loads. The column buckling bar (67.6) is capped for readability.

10. Conclusions and Recommendations

10.1 Conclusion on the deployed configuration

With the legs deployed in the # arrangement, the crane can be slewed, which it cannot do safely on its wheels. That is the real gain from putting the legs down. It does not make the crane compliant with its rated chart, and on Q235 the structural margin is thinner than the earlier S275 revisions showed.

Forward tipping is unchanged from the wheels-only case and, on the 980 mm front line, fails at mid-reach with a factor of 0.96, below one. Transverse legs add no forward reach, so they cannot help. Sideways tipping is governed by the deployed track. The 2500 mm track clears the target only close in and falls short at mid and far reach, where about 2784 mm is needed. The deployed leg member is sound at 1.64, but on the weaker steel the base outrigger arm now fails at 1.30, the boom outer fails at 1.26, the column sits at 0.85 and its base weld at 0.73.

So the crane cannot be certified to the full 1 tonne chart on a working-stress basis in the deployed configuration any more than on its wheels. The legs take care of sideways stability, provided the track is set wide enough, but they do not touch the forward-tipping shortfall or the structural items, and the Q235 material lowers the structural margin across the board.

10.2 The governing items in the deployed state

- Forward tipping, factor 0.96. Below one at mid-reach. Carried over from the wheels-only case and not improved by transverse legs or by the material. Resolved only by re-rating the chart or cutting reach.
- Sideways track, 2500 mm against 2784 mm needed. The one item the deployed configuration itself must fix. Confirm the deployed width and, if the feet adjust, set the track to at least 2784 mm.
- Column, factor 0.85, and column-base weld, factor 0.73. Carried over and now lower on Q235. The structural headline, addressed by the re-rate or upgrade.
- Boom outer and base outrigger arm, factors 1.26 and 1.30. Both fail on Q235 where they were marginal or passing on S275. The re-rate that fixes the column also brings these back above target.

10.3 Recommendations

Confirm the deployed footprint first. Measure the front foot position and the lateral track on the machine. If the legs angle forward at all, forward stability will be better than this report shows, and if the track opens past 2784 mm, sideways stability clears the target. Both turn on numbers the drawings give only in part.

Treat the structural and forward-tipping items through the re-rate. Re-rating the chart to what the Q235 steel and the stability safely carry clears every working-stress check in either configuration and needs no fabrication. Holding the full 1 tonne rating needs the larger column section and matching base weld. Deploying the legs is a precondition for slewing, not a substitute for that work.

Recommendation

Do not slew the crane until the deployed track is confirmed to at least 2784 mm. Do not work the crane to the current chart until forward tipping, the column, the base weld, the boom outer, the outrigger arm and the cylinder pin are addressed. Putting the legs down is required before any slew, but on its own it does not make the rated chart safe.

11. Assumptions and Items to Confirm

The leg parts and the section sizes come straight off the drawings. The material, the deployed positions, and a few inputs shared with the wheels-only report are confirmed, estimated or part-dimensioned and should be checked against the built machine before the report is relied on.

- Material, Q235. Client-confirmed. Nominal yield 235 MPa used for all verdicts; all sections are 6 mm wall, so no thickness de-rating applies. Mill certificates would confirm the actual yield, which commonly runs a little above nominal and would add margin.
- Deployed lateral track, 2500 mm. Read off the updated general-arrangement plan view. This is the make-or-break input for sideways stability; the as-built deployed width must be measured, and if the feet adjust, set to at least 2784 mm.
- Front foot ahead of column, 980 mm. From the base top-view chain and the warning-plate 420 mm dimension, which agree. The 2360 mm wheelbase on the new drawing is unchanged, so this holds. Worth a tape check on the machine.
- Leg deployment angle, 90 degrees. Taken from the deployment schematic. A forward splay would advance the front line and improve forward stability. The angle should be confirmed, since it drives both the forward foot position and the track.
- Leg arm, 750 mm, and section, RHS 60 x 100 x 6. From DETAIL-11 and the drawing. DETAIL B-B shows a 798 mm variant; confirm which is fitted.
- Crane self-weight, 700 kg, and CG offset, 0.30 m behind the column. Client-supplied and estimated respectively. A swing of plus or minus 100 kg moves both stability results, and the CG offset is the one input the drawings do not give, so it should be settled by weighing.

12. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- GB/T 700, carbon structural steels (Q235)
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

Appendix A: Companion files

- MATLAB R2026a parametric script (jib_crane_verification_C2_v3.m), with the material set to Q235 and the deployed track at 2500 mm
- Console log (console_log_C2_v3.txt), the full run including the deployed stability tables and the leg-member check
- Wheels-only and combined verification reports, the parent documents for the carried-over results
- Figures, grayscale, as used in this report